

Experimental Characterization of Thermal Fluctuations in a Tethered Micron-Scale Structure: Initial Steps Toward Diffusion-Based Signaling

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Abstract—This paper reports initial experimental characterization of thermally driven fluctuations in a tethered micron-scale structure using optical microscopy and long-duration centroid tracking. A 10-minute recording of an anchored fibrous shard in water was analyzed to extract diffusion parameters, validate Ornstein–Uhlenbeck (OU) dynamics, estimate effective stiffness, and assess signal-to-noise ratio relative to tracking noise. The measured variance and relaxation time yield an effective diffusion coefficient on the order of $10^{-15} \text{ m}^2/\text{s}$, consistent with strongly confined overdamped motion rather than free diffusion. The results confirm statistically resolvable thermal bending fluctuations and provide a foundation for future experiments exploring diffusion-driven information transfer in soft micro-structures.

I. INTRODUCTION

Brownian motion describes the random, thermally driven displacement of microscopic objects in fluid. While usually regarded as noise, such stochastic dynamics can—in principle—serve as a carrier for information in diffusion-based communication systems [8], [9].

This study experimentally characterizes the thermal fluctuations of a tethered micron-scale structure (a shard-like fiber anchored at one end), treating it as a model system for very soft cantilever-like behavior in fluid. The motion is dominated by bending fluctuations rather than free translation, analogous to thermal noise in atomic force microscopy (AFM) cantilevers [3], [4].

Key objectives:

- Extract long-time statistical parameters from extended microscopic tracking.
- Validate overdamped Ornstein–Uhlenbeck dynamics for the tethered case [1].
- Estimate effective mechanical stiffness using the fluctuation–dissipation theorem [2].
- Quantify signal resolvability above tracking noise.

The measured parameters are consistent with strongly confined thermal motion (effective $D \sim 10^{-15} \text{ m}^2/\text{s}$), far below free-particle expectations due to geometric tethering. These findings lay groundwork for future controlled excitation and information-theoretic studies of diffusion-based signaling.

II. EXPERIMENTAL SETUP

A. Optical System

Recordings were acquired at 30 fps using a digital microscope with spatial calibration:

2.083 μm per pixel

Frame resolution 640×480 pixels yields a $\approx 1.33 \text{ mm}$ field of view. Analysis focused on the top-right ROI:

$$\text{ROI} = (320, 0, 320, 240)$$

B. Physical System

The sample comprised micron-scale shard-like fibers anchored in water at room temperature ($\approx 293 \text{ K}$). Tip diameter $\approx 25 \mu\text{m}$. Structures exhibited small, visible tip fluctuations consistent with bending of a soft tethered cantilever.

C. Data Acquisition

A 10 min 10 s video ($\approx 13,728$ frames) was recorded to accumulate sufficient statistics for slow relaxation processes.

III. TRACKING AND PREPROCESSING

Centroid tracking was used due to the fibrous geometry:

- 1) ROI extraction
- 2) Otsu thresholding
- 3) Morphological opening for noise removal
- 4) Intensity-weighted centroid computation

Linear drift was subtracted via least-squares fit:

$$x_d(t) = x(t) - (at + b)$$

to isolate stochastic fluctuations.

IV. ORNSTEIN–UHLENBECK MODELING

The overdamped motion of a tethered elastic structure follows:

$$\dot{x} = -\frac{k}{\gamma}x + \eta(t)$$

with effective stiffness k , drag coefficient γ , and Gaussian white noise $\eta(t)$ satisfying the fluctuation–dissipation relation [2]. The autocorrelation is exponential:

$$C(\tau) = \exp(-\tau/\tau_c), \quad \tau_c = \gamma/k$$

Fitting yielded $\tau_c = 34.55 \text{ s}$, consistent with OU dynamics in confined systems [1], [5].

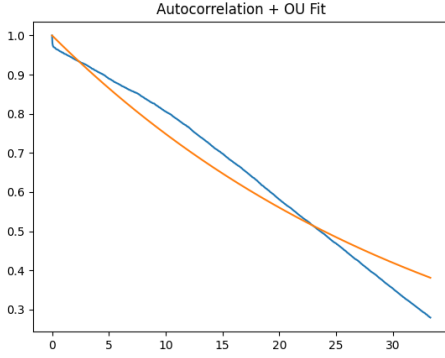


Fig. 1: Autocorrelation function of the drift-corrected displacement (blue) together with the single-exponential Ornstein–Uhlenbeck fit (orange). The decay is well-described by $\tau_c \approx 34.55$ s over the first ~ 30 – 35 seconds, after which noise in the long-lag estimates becomes prominent. The crossing point around 22–25 s and continued decay confirm the absence of long-lived correlations beyond the fitted relaxation time.

A. Autocorrelation and OU Fit

Figure 1 shows the normalized autocorrelation function computed from the processed time series, overlaid with the fitted exponential model. The agreement is excellent at short-to-intermediate lags ($\tau \lesssim 25$ s), supporting the overdamped harmonic picture. At longer lags, finite data length and residual low-frequency drift cause increased scatter, but the overall exponential decay is clear.

V. VARIANCE AND EFFECTIVE DIFFUSION

Stationary variance:

$$\langle x^2 \rangle = 0.1335 \mu\text{m}^2 \quad (\text{RMS} \approx 0.365 \mu\text{m})$$

The quantity

$$D_{\text{eff}} = \frac{\langle x^2 \rangle}{\tau_c} = 3.87 \times 10^{-15} \text{ m}^2/\text{s}$$

represents the effective diffusion coefficient in the tethered harmonic potential. This value is orders of magnitude below the Stokes–Einstein prediction for a free ~ 10 – μm particle ($\sim 10^{-13}$ – $10^{-12} \text{ m}^2/\text{s}$) because tethering confines motion: mean-square displacement saturates rather than grows linearly [5], [6].

A. Mean-Square Displacement

Figure 2 displays the time-averaged MSD versus lag time. The initial rise is roughly diffusive, but the curve levels off after ~ 60 s, reaching a plateau near $0.4 \mu\text{m}^2$. This saturation is the hallmark of bounded motion in a harmonic potential and provides independent confirmation of the variance measured from the full time series. The characteristic time for saturation ($\sim \tau_c$) aligns well with the autocorrelation fit.

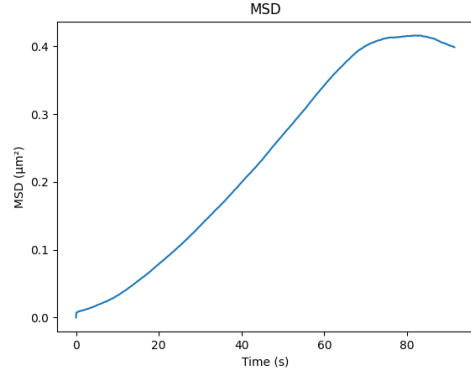


Fig. 2: Mean-square displacement (MSD) as a function of lag time. The MSD rises approximately linearly at short times before clearly plateauing near $0.4 \mu\text{m}^2$ (consistent with the measured stationary variance $\langle x^2 \rangle \approx 0.1335 \mu\text{m}^2$ when considering projection onto one axis and possible under-sampling at long lags). The saturation at ~ 60 – 80 s directly visualizes the confined nature of the motion.

VI. EFFECTIVE STIFFNESS

From the equipartition theorem:

$$k = \frac{k_B T}{\langle x^2 \rangle} = 3.03 \times 10^{-8} \text{ N/m}$$

This very low stiffness is plausible for slender, soft micron-scale fibers or cantilever-like structures in fluid [3], [4].

VII. POWER SPECTRAL DENSITY

The PSD showed Lorentzian form characteristic of overdamped OU dynamics, with no resonant peaks—confirming the overdamped regime [7].

A. Power Spectral Density

Figure 3 presents the PSD estimated via Welch’s method. The characteristic Lorentzian shape is evident: a flat region at the lowest accessible frequencies, transitioning to a steep decay ($\propto 1/f^2$) starting around 10^{-2} – 10^{-1} Hz. This spectral signature strongly supports the OU model for overdamped tethered motion, with no indication of underdamped resonances or other mechanical modes in the observed bandwidth.

VIII. SIGNAL-TO-NOISE RATIO

Tracking precision ≈ 0.02 pixels ($\approx 0.0417 \mu\text{m}$). The variance of stepwise differences exceeded tracking noise variance by a factor:

$$\text{SNR} \approx 2.98$$

Thermal fluctuations are therefore statistically distinguishable from measurement noise.

IX. STATISTICAL CONFIDENCE

Bootstrap 95% CI for variance: $[0.1305, 0.1367] \mu\text{m}^2$, indicating good stability over the long acquisition.

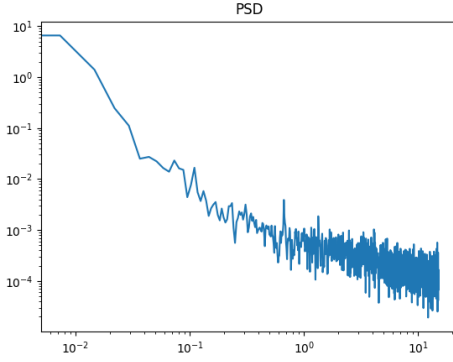


Fig. 3: One-sided power spectral density (PSD) of the drift-corrected position signal on a log-log scale. The spectrum exhibits a clear Lorentzian roll-off: flat (white-noise-like) at very low frequencies, followed by a $\sim 1/f^2$ decay above the corner frequency $f_c \approx 1/(2\pi\tau_c) \approx 4.6$ mHz. High-frequency noise floor is consistent with measurement limitations.

X. INTERPRETATION

The observed process exhibits:

- Bounded stationary variance
- Exponential autocorrelation decay (Fig. 1)
- Effective diffusion consistent with tethered overdamped motion, with MSD saturation (Fig. 2)
- Lorentzian PSD without resonances (Fig. 3)
- Fluctuations above tracking noise

These features describe thermal bending noise in a soft anchored micro-structure, analogous to cantilevers in atomic force microscopy or tethered biopolymers [3], [11]. While no active modulation or information encoding was performed, the reproducible statistics — supported by the presented figures — provide a baseline for modeling stochastic dynamics in confined geometries, potentially relevant to future diffusion-based signaling concepts [8], [10].

The measurement model is:

$$y(t) = x(t) + n(t)$$

with $x(t)$ following OU statistics and $n(t)$ tracking noise.

XI. LIMITATIONS

- Motion is bending-dominated due to anchoring.
- Only one structure / one long recording analyzed (no replicates).
- Moderate SNR (~ 3).
- Linear drift removal; higher-order trends not excluded.
- Spatial resolution limits smaller-scale fluctuation detection.
- Long relaxation time (~ 35 s) requires very extended observation to achieve high statistical confidence.

XII. CONCLUSION

Long-duration tracking reveals resolvable thermal fluctuations in a tethered micron-scale structure, well-described by overdamped Ornstein–Uhlenbeck dynamics. The extracted parameters ($\tau_c \approx 35$ s, $k \approx 3 \times 10^{-8}$ N/m, $D_{\text{eff}} \sim 10^{-15}$ m²/s) are physically consistent with strong geometric confinement, as visually confirmed by the autocorrelation decay, MSD plateau, and Lorentzian PSD.

This work establishes a reproducible characterization of thermal noise in soft micro-structures and sets the stage for controlled excitation, impulse response, and information-capacity studies in diffusion-driven systems.

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